A REGULAR ALTERATION FORCES A FINITE NORMALIZATION AND A REGULAR DENSE OPEN

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Editorial note. The alteration definition supplies finiteness of f over a nonempty open; shrinking preserves density since Y is integral.

Allowed standard prerequisites (editorial).

- Global sections of a proper space are finite over the base. Cohomology of Spaces, lemma proper-pushforward-coherent (cross-chapter, named). Makes C = Gamma(X, O_X) a finite A-module.
- Sections of a normal proper space form a normal domain. Spaces over Fields, lemma normal-integral-sections (cross-chapter, named). Shows C is a normal domain, so the integral closure B of A lands in C.
- Generic flatness. Morphisms of Spaces, proposition generic-flatness-reduced (cross-chapter, named). Makes f^{-1}V -> V flat after shrinking V.
- Faithfully flat descent of regularity. Algebra, lemma descent-regular (cross-chapter, named). Concludes V is regular from the faithfully flat cover.

Source reference. Original reference 07NG: focused-seeds/geometry/S9-spres-regular-alteration-implies/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08AR: focused-seeds/geometry/S9-spres-regular-alteration-implies/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06QS: focused-seeds/geometry/S9-spres-regular-alteration-implies/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ADA: focused-seeds/geometry/S9-spres-regular-alteration-implies/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BH3: focused-seeds/geometry/S9-spres-regular-alteration-implies/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted

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Lemma 0.1. *Let S be a scheme. Let Y be a Noetherian integral algebraic space over S. Assume there exists an alteration $f : X \to Y$ with X regular. Then the normalization $Y^\nu \to Y$ is finite and Y has a dense open which is regular.*

Proof. By étale localization, it suffices to prove this when $Y = \mathrm{Spec}(A)$ where A is a Noetherian domain. Let B be the integral closure of A in its fraction field. Set $C = \Gamma(X, \mathcal{O}_X)$. By Cohomology of Spaces, Lemma 08AR we see that C is a finite A-module. As X is normal (Properties of Spaces, Lemma 0BGT) we see that C is normal domain (Spaces over Fields, Lemma 0BH3). Thus $B \subset C$ and we conclude that B is finite over A as A is Noetherian.

There exists a nonempty open $V \subset Y$ such that $f^{-1}V \to V$ is finite, see Spaces over Fields, Definition 0ADA. After shrinking V we may assume that $f^{-1}V \to V$ is flat (Morphisms of Spaces, Proposition 06QS). Thus $f^{-1}V \to V$ is faithfully flat. Then V is regular by Algebra, Lemma 07NG. □